Proposal outline: Data leakage in flight-delay prediction

1. Research question

- **Main question:** How much does information unavailable two hours before scheduled departure influence prediction of arrival delay?
- **Focus:** Investigate how using information unavailable at the prediction time changes model performance, comparing a clean predictor set with predefined leakage conditions.

2. Data and background

Background and motivation

- **Existing problem:** Why it matters? How is it connected to our study**
- **Relevant studies**

Data sources

- **BTS Reporting Carrier On-Time Performance:** Provides flight related information: flight dates, airlines, origin/destination airports, schedules, distance, arrival-delay outcomes, etc..
- **Iowa Environmental Mesonet (IEM) METAR/ASOS archive:** Provides airport-level weather observations: match wind speed and precipitation to corresponding flight locations and times.

Full-study scope

- **Locations:** U.S. domestic flights across airports availably represented in the BTS dataset, with weather matched at both origin and destination.
- **Development (training + validation) period:** January–June 2025, including chronological validation for selection and tuning.
- **Final evaluation (test) period:** July-September 2025, final testing only.
- **Eligible flights:** Completed, non-diverted flights with available observed outcome.

Data Structure

- **Outcome:** Whether arrival is at least 15 minutes late, represented by `ArrDel15`.
- **Prediction time (cutoff):** Two hours prior to the scheduled departure.
- **Leakage Variables:**
  - Weather leakage: wind measurements and accumulated precipitation amounts taken near the scheduled departure time.
  - Operational leakage: `DepDelay` and `TaxiOut`.
- **Candidate Predictors:**
  - Predictors from source datasets: scheduled departure time, scheduled duration, distance, airline, origin/destination, weekday, and weather measurements.
  - Derived predictors: holiday indicators, historical delay rates.

3. Proposed comparison and primary estimands

- **Comparison:** Fit both logistic regression and random forest models under three conditions:

Condition	Weather and operational information
Clean (C)	Origin&destination wind and precipitation available before the cutoff; operational information available before the cutoff
Weather leakage (W)	Add later wind measurements and accumulated precipitation amounts taken near the scheduled departure time, on the basis of model C
Weather + operational leakage (WO)	W plus `DepDelay` and `TaxiOut`

- **Primary metric:** Average Precision (AP): definition? Higher the better.
- **Secondary metric:** Brier score: definition? Lower the better.

4. Evaluation plan (NOT YET CHECKED!)

- **Development data:** January–June 2025. Use expanding chronological training windows followed by later validation periods.
- **Model development:** select predictors, assess a limited set of plausible interactions, and tune logistic regression and random forest using development data only. Learn preprocessing within each training window; standardize numerical predictors if using penalized logistic regression.
- **Final fitting:** freeze feature definitions, selected predictors, preprocessing, and tuning choices, then refit on the declared development data.
- **Final test:** July–September 2025. Reserve this period from model selection and tuning.
- **Paired comparison:** evaluate clean (C), weather leakage (W), and weather plus operational leakage (WO) on identical test flights. Use comparable tuning budgets and consistent missing-data handling.
- **Information timing:** clean features must be available before the two-hour prediction cutoff. Historical rates may update sequentially using earlier observable outcomes, without refitting or tuning.
- **Performance evaluation:** evaluate all model conditions using AP and Brier score, and estimate the paired performance differences defined in the estimand section.
- **Reference and reporting:** include a constant-probability reference estimated from training outcomes. Report pooled and monthly scores and delay prevalence. Month is an evaluation grouping, not a candidate predictor.
- **Uncertainty:** estimate paired confidence intervals by resampling test dates, considering longer date blocks if dependence is material.
- **Interpretation:** the current C-to-W comparison changes both weather timing and precipitation representation—early intensity versus later accumulated amount. It therefore does not isolate timing alone. W-to-WO holds weather fixed.

5. Minimum viable experiment

We conducted a preliminary experiment using flights departing from Atlanta in January 2025. Models were fitted on 14,093 flights from January 1 to 20 and evaluated on 3,636 validation flights from January 21 to 25. The test set contains 4,851 flights from January 26 to 31, which remain reserved and have not been scored. Weather observations were matched to the corresponding flights at the departure airport.

The preliminary validation results and their interpretation are provided in Appendix A. The completed data matching, model fitting, and evaluation support the feasibility of the proposed experiment.

The minimum viable experiment compares both models under the three information conditions using a sample limited in time and geographic coverage. The full study will expand coverage to IEM-covered airports across the United States and include weather conditions at destination airports.

6. Computing requirements and verified access to data and code

We will use R and the `ranger` package on a CPU-based laptop. Data will be downloaded and processed in a month unit.

Access to the required BTS monthly files has been verified, and IEM weather data have been verified for selected airports. The preliminary analysis has confirmed that flight and weather data can be combined for model fitting. Information availability, and computing requirements, including runtime and memory use, will be checked further upon the full study.

7. Fallback and responsibilities

If computing limits require sampling, we will sample proportionally within month, origin-airport, and airline groups. We will preserve natural delay prevalence and use identical flights across model conditions. Sample coverage will be checked, and weights will be applied if flights have unequal selection probabilities. Further fallbacks include shortening historical windows from 365 to 182 days, reducing tuning or tree budgets, and restricting airport coverage. Weather leakage remains central to the study; forecast data remain an option for later investigation.

Responsibilities will be divided as follows:

References to use

- STA2112, *Final Project Guidelines and Seed Questions*, Fall 2026, September 9, 2026; relevant lectures on legitimate information and evaluation.
- BTS field documentation.
- BTS monthly download archive.
- IEM METAR/ASOS observations and field definitions.
- IEM archived TAF forecasts: a future clean-weather option, not the current primary baseline.
- National Weather Service weather-product documentation: observation/forecast interpretation.
- U.S. federal holiday calendars.
- Chronological validation guidance.
- Statistics Canada sampling guidance.
- Average precision definition: methodological reference; the pilot calculates AP in R.
- Brier score definition: methodological reference for binary mean squared probability error; the pilot calculates Brier score in R.
- ranger documentation.
- glmnet documentation: penalized logistic regression, alpha/lambda tuning, and predictor standardization.
- Feature-selection and preprocessing leakage guidance: methodological guidance applicable to the R workflow.
- Optional model-background reference: Probst, Wright, and Boulesteix (2019), Hyperparameters and tuning strategies for random forest.

Appendix A. Preliminary pilot results

Table A1 presents the results of the untuned models on the same 3,636 validation flights departing from Atlanta from January 21 to 25, 2025. The January 26–31 test set was not scored.

Table A1. Pilot validation performance under three information conditions.

Model	Clean	Weather leakage only	Weather + operational leakage
Logistic regression	0.2291 / 0.2776	0.2261 / 0.2777	0.9764 / 0.0389
Random forest	0.3509 / 0.2570	0.4286 / 0.2421	0.9710 / 0.0414

Each cell reports AP / Brier, where higher AP and lower Brier indicate better performance.

The weather-leakage condition improved random forest performance but produced little change in logistic regression performance. Adding departure delay and taxi-out duration produced much larger apparent gains for both models.

The clean model uses early precipitation intensity, while the leakage models use later accumulated precipitation amounts. The weather comparison therefore changes both timing and precipitation representation and does not isolate timing alone. These results provide preliminary evidence from a single airport and a five-day validation period; they do not establish performance across the full study population.

Appendix: working notes and supporting evidence — exclude from the submitted two-page proposal

A. Predictor construction and information timing

- **Predictor groups used in the pilot:**
- Scheduled information: `DayOfWeek`, scheduled departure minutes, `CRSElapsedTime`, and `Distance`.
- Unavailable information at prediction time: `DepDelay` and `TaxiOut`, added to the same clean predictors.
- Never include `ArrDel15` or direct encodings of arrival delay as predictors.
- Explain the feature-availability assumptions using BTS documentation.
- The archive is retrospective; treating recorded scheduled fields as known at the cutoff is an assumption to acknowledge.
- Airline and destination are downloaded but are not predictors in the current pilot.
- **Candidate predictors and feature construction:**
- **Raw and transformed fields:** audited airline, origin/destination airport, route, scheduled times, duration, distance, and weekday.
- Remove constant and redundant representations before selection.
- **Calendar:** categorical month, a U.S. federal-holiday indicator, and a predefined surrounding-holiday window derived from flight dates and an official calendar.
- Specify actual versus observed holiday dates.
- Month represents calendar variation, not direct control of climate; plausible geographic interactions may be investigated during development.
- **Historical delay rates:** construct rolling **365-day** rates for airline, origin–destination route, and airline–flight-number–route groups, using only eligible outcomes observable before each prediction cutoff.
- Include history counts and prespecify smoothing or broader-group fallbacks for sparse or new services.
- Use **182 days** as a computational fallback.
- Complete histories require approximately January 2024–June 2026 records; these additional records support features, not extra model-fitting periods.
- **Clean weather:** latest origin and destination observations available by the common cutoff, two hours before scheduled departure.
- Convert times to UTC, use backward matching with a maximum report age, and audit publication latency/corrections; observation time alone does not prove availability.
- Document the condition-specific precipitation definitions, wind units, accumulation intervals, trace values, and missing-data rules.
- The current clean model uses categorical intensity; both leakage models use `p01i` accumulated amounts in inches.
- The ASOS `p01i` field is accumulated precipitation, potentially including frozen precipitation, not an instantaneous rain-only rate; document this limitation if used.

- **Forecast option:** whether forecasts or pre-cutoff measurements should be the eventual clean weather representation remains open.
- Use pre-cutoff METAR/ASOS measurements for the current proposal design.
- Reassess forecasts during post-proposal development, before final testing, rather than conclude that they are generally inferior from the pilot.
- Eligible forecasts must be available by the same cutoff; origin and destination target times would be scheduled departure and arrival respectively.

B. Comparison and metric implementation

- The current C-to-W comparison changes both weather timing and precipitation representation (clean intensity versus later `p01i` amount), so it does not isolate timing alone.
- W-to-WO holds weather fixed and adds operational information.
- A future timing-only sensitivity check would require the same precipitation representation at both times.
- Check that later reports occur after the prediction cutoff.
- Prespecify the destination leakage matching time before final evaluation, with scheduled arrival a candidate; actual arrival time is not a settled choice and would introduce an additional outcome-dependent matching mechanism.
- A secondary nested comparison may retain early weather and add later weather, but label it separately because it increases predictor count.
- For each paired feature comparison, use identical eligible flights, training dates, and evaluation dates.
- Before final model fitting and comparison, develop the clean models through variable selection and hyperparameter tuning using chronological development windows.
- Within each model, freeze selected non-weather predictors and prespecify condition-specific weather definitions before comparing conditions or adding operational leakage.
- Use comparable tuning rules and budgets across all three information conditions.
- Describe the latter condition as a constructed hindsight workflow; do not imply that it reproduces a published study unless a specific protocol is identified.
- **Primary metric:** Average precision (AP), computed using the same tie-aware function in `scripts/metrics.R` for every model.
- AP measures ranking and is not classification accuracy or trapezoidal PR-AUC.
- **Secondary metric:** Binary Brier score, calculated with the shared `calculate_brier_score(actual, probability)` helper in `scripts/metrics.R`: `mean((probability - actual)^2, na.rm = TRUE)`.
- Lower is better.
- It evaluates probability accuracy and reflects both calibration and discrimination, rather than calibration alone.
- Use numeric 0/1 outcomes and the same evaluation flights for every model.
- The simple helper omits missing squared errors; the current pilot has complete outcomes and probabilities, so its scores are unchanged.

- For future comparisons, use a common complete evaluation set.
- Log loss can remain an optional extension; it is not calculated in the current pilot.
- **Primary estimands, separately for each model:**
- Weather-information/representation contrast: `Delta_AP_weather = AP_W - AP_C`; do not attribute the full difference to timing alone.
- Additional operational leakage: `Delta_AP_operational = AP_WO - AP_W`.
- Total apparent gain: `Delta_AP_total = AP_WO - AP_C`.
- **Secondary estimands:** the corresponding Brier reductions, `Brier_C - Brier_W`, `Brier_W - Brier_WO`, and `Brier_C - Brier_WO`.
- Positive differences indicate improved apparent performance.
- These are paired predictive-performance contrasts, not causal estimates of weather effects.
- Report improvements or deterioration rather than assuming a monotonic increase across conditions.
- **Secondary comparison:** Does correcting information availability change the model ranking?
- Interpret small differences using uncertainty estimates.
- Boosting is optional and outside the minimum core study.

C. Detailed chronological development and evaluation protocol

Specify the design before final evaluation:

- **Development: January–June 2025.** Use expanding chronological windows for variable selection, interaction assessment, and tuning.
- For a categorical month predictor, include earlier observations from the validation month in training: e.g., train through March 15 and validate March 16–31, then expand through April 15 and validate April 16–30.
- Continue into May and June; prespecify folds and handling of unseen airport/airline levels.
- Each validation window follows its training data.
- **Final fitting:** freeze feature definitions, selected terms, weather matching, missing-data rules, and tuning choices, then refit on the declared January–June 2025 development sample.
- **Final evaluation: January–June 2026.** Reserve outcomes from model selection and tuning.
- Generate predictions chronologically at the two-hour cutoff; fitting data and any outcome-based training features must respect outcome-availability times.
- **Sequential history updates:** model parameters remain fixed, while historical-rate features may update using earlier outcomes once observable.
- This prospective simulation can use earlier 2026 outcomes for later histories, but never the current/future outcome or test-based retuning.
- Data in intervening months are needed to maintain full rolling histories.

Include:

- **Variable selection:** Compare the full audited clean pool with parsimonious predictor-group subsets.
- Learn data-driven selection inside each training window and use its subsequent validation period to choose subsets; never select using the final test outcomes.

- **Interaction investigation:** assess a limited, flight-operation-motivated set, such as month × location/region or holiday timing × location.
- Logistic regression requires explicit candidate interaction terms, retaining their main effects; random forest can learn interactions through tree splits.
- Selection alone does not automatically generate logistic interactions.
- Evaluate these choices inside development windows rather than searching all combinations.
- **Scaling:** Standardize numerical predictors for penalized logistic regression using each training window's means and standard deviations, applied unchanged to its validation data.
- Encode categorical variables appropriately.
- Scaling is not generally required for the current unpenalized logistic model or random forest; random-forest predictors will retain their original numerical scales.
- **Logistic tuning:** Retain the existing unpenalized logistic regression as a reference and consider a lasso/elastic-net extension, tuning penalty strength (lambda) and mixing (alpha) over a bounded grid.
- Ordinary glm() coefficient estimation is model fitting, not hyperparameter tuning.
- **Random-forest tuning:** Tune predictors considered at each split (mtry), node size, and possibly tree depth; bound the number of trees for feasible computation.
- Record candidate settings, seeds, and budgets.
- Use AP as the primary selection criterion and report Brier score as a secondary diagnostic.
- Predefine a tie rule and avoid choosing whichever metric happens to favour a candidate.
- Fit all learned preprocessing, including scaling and selection, within the training portion of each chronological window.
- After choices are frozen, refit on the declared development data and apply the resulting transformations to the untouched final period.
- Evaluate all three conditions on exactly the same final flights.
- Prespecify missing-weather handling rather than dropping different flights by condition; estimate learned imputations from training data only.
- Report pooled and monthly delay prevalence, AP, and Brier score.
- Pooled AP must be computed from pooled predictions, not by averaging monthly AP.
- Report the same metrics for development windows.
- Use a constant-probability reference estimated from the training outcomes only; do not fit the reference probability to validation or test outcomes.
- Estimate uncertainty in AP and Brier-score differences with paired resampling of test dates.
- State that this addresses variation across evaluation dates, conditional on fitted models; it does not capture all training uncertainty.
- Consider longer blocks if dependence across dates is material.
- Use additional chronological development windows to assess sensitivity to time periods, keeping final evaluation separate.
- If retained, random-split evaluation is a separate diagnostic.
- Do not combine feature removal and split changes into one comparison and attribute the full difference to either factor.

D. Supporting pilot tables and interpretation

- **Completed pilot:** January 2025 Atlanta departures; 22,580 eligible flights.
- All training and validation flights remain under the shared complete-case mask.

Pilot set	Dates	Flights	Delay rate
Training	January 1–20	14,093	24.52%
Validation	January 21–25	3,636	33.55%
Reserved pilot test	January 26–31	4,851	11.79%

- **Current weather definitions:** origin wind in knots for all conditions.
- Clean precipitation remains unordered METAR intensity (none/light/moderate/heavy), including snow and other supported falling precipitation.
- For both leakage conditions, **Precipitation amount:** IEM's `p01i`, measured in inches accumulated since the previous hourly precipitation reset—not categorical intensity.
- Special-report accumulation windows can be shorter than an hour; the amount can include frozen precipitation and is not an instantaneous rain-only rate.
- Trace values retain the existing 0.0001-inch encoding, not a measured trace amount.
- C uses early wind/intensity; W uses later wind/amount near scheduled departure; WO adds `DepDelay` and `TaxiOut` to W.
- The same non-weather predictors and flight rows are used throughout, but C-to-W changes precipitation representation as well as timing.
- W-to-WO holds weather fixed.

Model	Condition	AP (higher is better)	Brier (lower is better)
Constant reference	Training delay prevalence	0.3355	0.2311
Logistic regression	Clean early weather (C)	0.2291	0.2776
Logistic regression	Later weather (W)	0.2261	0.2777
Logistic regression	Later weather + operational leakage (WO)	0.9764	0.0389
Random forest	Clean early weather (C)	0.3509	0.2570
Random forest	Later weather (W)	0.4286	0.2421
Random forest	Later weather + operational leakage (WO)	0.9710	0.0414

- **Paired apparent performance changes:** positive values indicate better apparent performance; negative values indicate deterioration.

Model	Comparison	AP gain	Brier reduction
Logistic regression	clean -> weather_leakage	-0.0031	-0.0001
Logistic regression	weather_leakage -> leakage	+0.7503	+0.2388
Logistic regression	clean -> leakage	+0.7472	+0.2387
Random forest	clean -> weather_leakage	+0.0777	+0.0148
Random forest	weather_leakage -> leakage	+0.5424	+0.2007
Random forest	clean -> leakage	+0.6201	+0.2155

- The clean random forest exceeds the constant reference on AP (0.3509 versus 0.3355), while clean logistic regression does not (0.2291).
- Both clean models have larger Brier error than the constant reference (0.2311).
- The later-amount weather condition improves random forest AP and Brier, but slightly worsens both metrics for logistic regression.
- Operational leakage gives a much larger gain.
- The C-to-W difference cannot be attributed solely to timing because the precipitation representation changes too; none of these results estimates a causal weather effect.
- Origin-only integration is verified; destination and nationwide matching remain later work.
- Later reports were 0–59 minutes before scheduled departure and 61–120 minutes after cutoff.
- Historical publication/receipt times and corrections remain availability assumptions.
- Do not mix the current mixed-representation results with earlier all-intensity, all-amount, or forecast experiments.
- The pilot test was not predicted or scored.
- The minimum experiment uses a bounded sample, both model families, and three conditions.
- The intended full study expands to nationwide origin and destination weather and paired uncertainty.
- Keep only a brief pilot summary in the submitted two-page proposal; these detailed tables are supporting material.
- **Reproducible workflow:** `source("scripts/run_pilot.R")` from the RStudio project root.
- To regenerate tables without refitting, use `source("scripts/04_proposal_results.R")`.
- **Current proposal artifacts:**
- `results/pilot/proposal_comparison.md`: readable comparison.
- `results/pilot/proposal_validation_results.csv`: full-precision AP/Brier scores.
- `results/pilot/proposal_performance_changes.csv`: paired changes.
- `results/pilot/proposal_validation_predictions.csv`: common flight IDs and probabilities.
- `results/pilot/proposal_models.rds`: both model families and fitted specifications.
- Earlier scripts, diagnostics, and result files are preserved under the dated `archive/proposal_cleanup_*` directory and are not required for the current proposal workflow.

E. Detailed access, computing, and task allocation

- **Access already verified:**
- January 2025 data downloaded and processed.
- Logistic-regression and random-forest scripts ran in the team's RStudio project.
- Shared AP calculation and saved model/result files are available.
- IEM Atlanta observation downloads for January–June 2025 and 2026 and pilot weather/model joins succeeded; pilot archived TAF access also succeeded.
- Nationwide destination coverage has not yet been verified.
- The repository now contains the integrated observed-weather pilot workflow, paired AP/Brier tables, fitted models, and shared validation predictions.
- Separate forecast experiments remain development options outside this primary workflow.
- **Still to verify:** remaining BTS development, evaluation, and 365-day history files; nationwide origin/destination weather coverage; airport–station/timezone mapping; report availability and correction handling; missingness; destination leakage matching time; package versions; execution on teammates' computers; full-study runtime and memory.
- **Computing plan:** R on an available CPU laptop, using base R and `ranger`; `glmnet` is a planned dependency if the penalized logistic extension is adopted and its installation must be verified.
- Use bounded selection and tuning searches.
- Record versions, seeds, runtime, and relevant resource limits.
- Process historical records and station reports in batches, retaining only needed fields and compact group summaries; calculate historical rates from full available past records before sampling modeling rows.
- Cache airport weather once and reuse it across flights.
- Benchmark time and memory before fixing sample size.
- No GPU or paid API is planned.
- **Sampling and fallback:** first reduce model-fitting cost through proportional random sampling within month × origin airport × airline groups, after chronological periods are defined.
- Preserve natural delay prevalence and identical sampled flights across conditions.
- Prefer full or fixed representative evaluation samples with batched prediction; use inclusion-probability weights for population evaluation and appropriate training weights if rare groups are deliberately oversampled.
- Check sample composition and development-seed sensitivity.
- If necessary, shorten histories from 365 to 182 days, reduce tuning/tree budgets, or declare a narrower airport scope.
- Weather leakage remains a core investigation; forecasts and extra aircraft-history joins are optional extensions.
- **Responsibilities:**

Name	Part	Detail
–NAME–	Data Cleaning & Preprocessing	Audit eligibility and information timing; build holiday/history features, airport–station/timezone mappings, and origin/destination weather joins; document sampling and missingness.
–NAME–	Model Training - Logistic Regression & Random Forest	Audit candidate subsets with the preprocessing member, assess plausible interactions, perform chronological selection/tuning, and fit the three conditions under documented budgets.
–NAME–	Evaluation & Inference	Prespecify paired AP/Brier comparisons and uncertainty, evaluate January–June 2026 after choices are frozen, and report pooled/monthly results and limitations without test-based tuning.